

The Effects of a 90-km Outdoor Cycling Ride on Performance Outcomes Derived From Ramp-Incremental and 3-Minute All-Out Tests

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Abstract

Bitel, M, Keir, DA, Grossman, K, Barnes, M, Murias, JM, and Belfry, GR. The effects of a 90-km outdoor cycling ride on performance outcomes derived from ramp-incremental and 3-minute all-out tests. *J Strength Cond Res* 38(3): 540–548, 2024—The purpose of this study was to determine whether laboratory-derived exercise intensity and performance demarcations are altered after prolonged outdoor cycling. Male recreational cyclists ($n = 10$; RIDE) performed an exhaustive ramp-incremental test (RAMP) and a 3-minute all-out test (3MT) on a cycle ergometer before and after a 90-km cycling ride. RAMP-derived maximal oxygen uptake ($\dot{V}O_{2max}$), gas exchange threshold (GET), respiratory compensation point (RCP), and associated power output (PO), as well as 3MT-derived critical power (CP) and work performed above CP, were compared before and after ~3 hours of outdoor cycling. Six active men served as “no-exercise” healthy controls (CON), who, instead, rested for 3 hours between repeated RAMP and 3MT tests. During the 90-km ride, the duration within the moderate-intensity, heavy-intensity, and severe-intensity domains was $59 \pm 24\%$, $40 \pm 24\%$, and $1 \pm 1\%$, respectively. Compared with pre-90 km, post-RAMP exhibited reductions in (a) $\dot{V}O_{2max}$ (4.04 ± 0.48 vs. 3.80 ± 0.38 L·min⁻¹; $p = 0.026$) and associated PO (392 ± 30 W vs. 357 ± 26 W; $p = 0.002$); (b) the $\dot{V}O_2$ and PO at RCP (3.49 ± 0.46 vs. 3.34 ± 0.43 L·min⁻¹; $p = 0.040$ and 312 ± 40 W vs. 292 ± 24 W; $p = 0.023$); and (c) the PO (214 ± 32 W vs. 198 ± 25 W; $p = 0.027$), but not the $\dot{V}O_2$ at GET (2.52 ± 0.44 vs. 2.44 ± 0.38 L·min⁻¹; $p = 0.388$). Pre-90 km vs. post-90 km 3MT variables showed reduced W' (9.8 ± 3.4 vs. 6.8 ± 2.6 kJ; $p = 0.002$) and unchanged CP (304 ± 26 W and 297 ± 34 W; $p = 0.275$). In the CON group, there were no differences in $\dot{V}O_{2max}$, GET, RCP, W' , CP, or associated power outputs ($p > 0.05$) pre-to-post 3 hours of rest. The preservation of critical power demonstrates that longer-duration maximal efforts may be sustained after long-duration cycle. However, shorter sprints and higher-intensity efforts eliciting $\dot{V}O_{2max}$ will exhibit decreased PO after 3 hours of a predominantly moderate-intensity cycle.

Key Words: long-duration exercise, cycling on the open road, $\dot{V}O_{2max}$, peak power, anaerobic capacity

Introduction

Trained cyclists and triathletes participate regularly in cycle training sessions from 20 to 200 km, including exercise intensities across the continuum from below the lactate threshold to the severe-intensity domain (26). Little is known about how much exercise duration, predominantly sublactate threshold exercise, affects laboratory-based or field-based measures of exercise performance (2). These measures include critical power (CP), work performed above CP termed anaerobic capacity (W'), and peak power output (PO). These can all be determined from an all-out effort 3-minute cycle performance (8,41). A hyperbolic relationship is observed when PO is plotted as a function of time to exhaustion within this 3-minute severe-intensity exercise (29). The power asymptote represents the CP and the work performed above this CP over the 3 minutes gives the fixed amount of work (kJ) that can be performed at any PO above CP (W'). CP theoretically represents the highest PO that can be maintained for a prolonged period without relying on nonoxidative energy sources, whereas the work performed above CP reflects anaerobic capacity.

In addition to CP and W' , the gas exchange threshold (GET), respiratory compensation point (RCP), and maximal oxygen consumption ($\dot{V}O_{2max}$) and associated incremental test-derived POs provide estimates of the discrete boundaries defining moderate-intensity, heavy-intensity, and severe-intensity exercise domains (23,24,44). However, it has not been investigated whether these metabolic boundaries and the PO at which they occur are affected by long-duration, predominantly moderate-intensity cycling.

Clark et al. (11) studied the effects of 2 hours of cycling at a heavy-intensity exercise in a laboratory-controlled environment on CP and W' as determined from the aforementioned 3-minute all-out test (13). Both PO at CP and W' were reduced post-exercise, on average, by 8 and 20%, respectively. The authors attributed the decline in CP and W' to intramuscular glycogen depletion and fatigue-inducing metabolite accumulation (11,37). Although these findings contest the assumption that the parameters of the power-time relationship remain fixed throughout prolonged exercise below CP, the 2-hour cycling in that study was performed within the heavy-intensity domain only. Their PO was carefully controlled and was different from a cycle ride experienced on a “real-world” ride where fluids would be ingested ad libitum and intensities may vary. Furthermore, for recreational and highly trained cyclists, durations over 2 hours are standard in

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training (12) and training is performed predominately within the moderate-intensity domain with intermittent excursions into the heavy-intensity and severe-intensity domains depending on terrain, head or tail winds, and ride strategy. Whether CP and W' would be reduced similarly after a prolonged (3 hours) outdoor cycling session predominantly performed within the moderate-intensity domain remains unknown.

This study examined how commonly used laboratory-derived physiological demarcations of exercise intensity and performance, measured immediately before a training session, are influenced by a long-duration, non-laboratory-based ("real-world") cycling session predominantly performed within the moderate-intensity domain. Accordingly, we hypothesized that the boundaries of the moderate-intensity (i.e., <GET), heavy-intensity (i.e., >GET—CP and RCP), and severe-intensity (i.e., $\dot{V}O_{2\max}$) domains and W' would be reduced after 90 km of predominantly moderate-intensity cycling. By contrast, these parameters would not change when 90 km of cycling was substituted with a 3-hour sedentary period.

Methods

Experimental Approach to the Problem

The control group (CON) was instructed to remain sedentary for 3 hours, and the long-duration cycle group (RIDE) arrived at the laboratory having avoided exercise and refrained from caffeine and alcohol consumption for a minimum of 24 hours. In the case of the CON group, subjects were instructed to consume a breakfast meal that they would typically ingest before a long-duration cycling session or heading off to work for the day. During the 90-km cycling, RIDE subjects were asked to consume their usual supplementary drinks and snacks for a long-duration ride. Breakfast and intervention food intake were recorded for both RIDE and CON groups. On arrival at the laboratory, both groups performed a ramp-incremental exercise test to task failure, defined as the inability to continue cycling or a drop in the pedal cadence <60 for more than 5 consecutive seconds despite strong verbal encouragement. This test is similar to cyclists navigating a climb several kilometers in length in which the intensity of the group continues to increase, a common experience in cycling. After a 15-minute period of a light-intensity recovery, a 3-minute all-out sprint test was performed. The 15 minutes of the lower-intensity recovery period simulated those periods within races or training sessions where intensity is low, whereas the 3-minute all-out test reflects a sustained sprint—as a planned part of the training session or in race situations where cyclists chase down an attack or attempt to break away from their competitors. Within 6 minutes of completion of the pre-ride testing, the RIDE group embarked on a 90-km outdoor cycling route (~3 hours). Similarly, within 6 minutes of completion of the 90-km cycle (RIDE) or after 3 hours of seated rest (CON), both groups repeated the ramp-incremental and 3-minute all-out test protocols.

Subjects

Sixteen individuals volunteered to participate in the study. The experimental group (RIDE) consisted of 10 male recreationally active cyclists (age range: 20-38 years; 28 ± 6 years; 75 ± 5 kg; 181 ± 7 cm; who have cycled $200 \text{ km} \cdot \text{wk}^{-1}$ for a minimum of 3 years) who were experienced in various maximal exertional efforts similar to the current study's testing procedures. The control (CON) group consisted of 6 age-matched, body mass-matched, and height-

matched healthy men (age range: 22-32 years; 27 ± 4 years; 78 ± 8 kg; 179 ± 8 cm) with previous experience with the current study's tests. In addition, the CON group consisted of 3 soccer players, 2 cyclists, and 1 resistance-trained individual. All performed at the club or intercollegiate level.

All subjects were nonsmokers with no known history of cardiovascular, respiratory, metabolic, or musculoskeletal conditions, nor were they taking medication that would be anticipated to affect their physiological responses to the study protocol. The study design and procedures were by the ethical standards of the Declaration of Helsinki and approved by the University of Western Ontario's Human Research Ethics Board. All subjects were informed of the risks and benefits of the study before any data collection and then signed an institutionally approved informed consent document.

Procedure

90-km Ride. Before testing, each subject in the RIDE group was familiarized with the 90-km cycling route with an elevation change of 330 meters. Subjects were outfitted with a Garmin FORERUNNER 310XT watch (Garmin, Kansas City, MO) that continuously acquired and stored heart rate (HR), time, and speed data. Throughout the 90-km ride, subjects were instructed to maintain a rate of perceived exertion of 11 on the 20-point scale (6) which is indicative of the upper limit of moderate-intensity domain exercise (35). All subjects completed the course on a clear and dry day using their road bikes.

Exercise Protocols. All laboratory-based exercise testing protocols were performed on an electromagnetically braked cycle ergometer (Velotron, RacerMate). The ramp-incremental test consisted of a 4-minute, 20 W baseline, followed by a linear rise in PO with a slope of $30 \text{ W} \cdot \text{min}^{-1}$. Subjects were instructed to maintain a cadence of 70 rpm throughout the protocol and to continue pedaling until exhaustion. Measurements of $\dot{V}O_{2\max}$, GET, RCP, and peak aerobic power achieved at task failure were derived from the ramp-incremental test. Under well-controlled conditions, subjects are known to reach $\dot{V}O_{2\max}$ at the end of a ramp-incremental test to task failure (19,30). To increase confidence that the efforts were maximal, we relied on secondary criteria: respiratory exchange ratio (RER) > 1.10, HR 90% of their age-predicted maximum ($\{0.90 \times (208 - [0.7 \times \text{age}])\}$), and their limit of tolerance (22,33). A 15-minute light-intensity recovery period was then performed. After the recovery period, the 3-minute all-out test was preceded by 40 seconds of 20 W cycling. At ~35 seconds of this 20-W cycling period, subjects were instructed to increase their cadence to 100 rpm. The resistive load was computed using a torque factor equivalent to the PO, approximating 50% of the difference between GET and $\dot{V}O_{2\max}$ as determined from the previous ramp-incremental test (41). With the cycle ergometer set in linear mode (linear factor = $\text{power} \cdot \text{preferred cadence}^{-2}$), at the 40-second mark, the resistive load was applied, and the subjects were then instructed to attain a maximal effort as quickly as possible and to maintain that effort throughout the 180-second test. Subjects were not informed of elapsed time to prevent pacing, and verbal encouragement was provided. The 3-minute all-out test provided CP and W' (8,41). Within 6 minutes of completion of the pre-ride testing, the RIDE group embarked on a 90-km outdoor cycling route (~3 hours). Similarly, within 6 minutes of completion of the 90-km cycle (RIDE) or within 6 minutes after 3 hours of seated rest (CON), both groups repeated the ramp-incremental and 3-minute all-out test protocols.

Data Collection. As previously described, gas exchange and ventilatory data were measured breath-by-breath during all laboratory-based exercise tests (25). In brief, the inspired and expired flow rates were analyzed by a low dead space (90 ml) bidirectional turbine (Alpha Technologies VMM 110). The turbine was calibrated before each test using a 3-L syringe. Inspired and expired gases were monitored at the mouth and analyzed for concentrations of O_2 , CO_2 , and N_2 by a mass spectrometer (Innovision, Amis 2000, Lindvedvej, Denmark). Flow volume and gas concentration data were then transmitted to a laboratory computer. The laboratory computer built a profile of each breath by aligning the gas concentration information with the inspiratory and expiratory volume recordings. Breath-by-breath gas exchange was determined from the algorithms developed by Swanson (38). Power output and HR (through Polar Wearlink Chest Strap, Kempele, Finland) data were recorded continuously on a second-by-second basis. Blood lactate was measured 3 minutes before and after exercise cessation for all exercise tests. Capillary blood samples were drawn from a pinprick of the index finger and immediately analyzed for lactate ($[La^-]$; mM) using a handheld lactate analyzer (Lactate Scout, SensLab GmbH, Germany).

Data Analysis. Breath-by-breath $\dot{V}O_2$ data for each exercise protocol were edited by removing aberrant data points lying 3 SDs outside the local mean, linearly interpolated to 1-second intervals, and further averaged into 5-second time bins (27). For both the ramp-incremental and 3-minute all-out tests, $\dot{V}O_{2\max}$ was identified as the highest 15-second average value.

To determine the GET and RCP, the profiles of ventilation ($\dot{V}_E$), gas exchange ($\dot{V}O_2$ and $\dot{V}CO_2$), their combination ($\dot{V}_E/\dot{V}CO_2$ and $\dot{V}_E/\dot{V}O_2$), and end-tidal partial pressures of O_2 and CO_2 were plotted against $\dot{V}O_2$. The GET was identified as the $\dot{V}O_2$ at which $\dot{V}CO_2$ and $\dot{V}_E$ began to increase disproportionately in relation to $\dot{V}O_2$, with a systematic rise in end-tidal PO_2 , whereas end-tidal PCO_2 and $\dot{V}_E/\dot{V}CO_2$ were stable. The RCP was identified as the $\dot{V}O_2$ at which end-tidal PCO_2 began to fall after an isocapnic buffering period, corroborated by a second break point in the $\dot{V}_E$ vs. $\dot{V}O_2$ relationship and a sudden rise in the $\dot{V}_E/\dot{V}CO_2$ vs. $\dot{V}O_2$ profile (45). Three blinded investigators independently identified the GET and RCP. If a discrepancy arose between the 3 investigators, a mean of the identified points was used as long as all estimates were within $200 \text{ ml} \cdot \text{min}^{-1}$. If one of the investigators' estimates was not within $200 \text{ ml} \cdot \text{min}^{-1}$, an average of the 2 in the closest agreement was used.

Power output at each threshold was determined by noting the PO at the $\dot{V}O_2$ from the established GET and RCP thresholds. This process also identified the HR corresponding to GET and RCP. In addition, to account for the time delay from the change in work rate to the change in $\dot{V}O_2$ during ramp-incremental exercise, each associated PO for GET and RCP was decreased by two-thirds of the ramp rate (20 W) (4). $\dot{V}CO_2/\dot{V}O_2$ determined the RER. The estimated CP was calculated as the mean PO over the last 30 seconds of the 3-minute all-out test, whereas W' was calculated as the power-time integral part above the CP area (41).

The total time spent within the 3 exercise intensity domains during the 90-km ride was estimated for each subject post hoc using their continuous HR data. The HRs corresponding to GET and RCP from the pre-ride and post-ride ramp-incremental test were used to establish the upper boundaries of the moderate-intensity and heavy-intensity domains from the beginning to completion of the 90-km ride. We assumed there was a linear change in the HR at GET over the duration of the ride. There were

only 2 heartbeat differences at GET between the mean pre-ride and post-ride ramp-incremental exercise (pre-HR 139; post-ramp-incremental exercise HR 141). This difference was not significant ($p = 0.661$). With these boundaries established, the continuous HR data were analyzed individually to estimate the time spent in each domain and as a percentage of total time with the assumption that there was a linear change in HR at GET pre-90 km to post-90 km of cycling.

Statistical Analyses

Statistical analyses used SigmaPlot version 12.3 (Systat Software Inc., San Jose, CA). A paired t -test analysis was used to compare measurements for possible differences before and after the intervention for both groups. A 2-way (group $\times$ time) analysis of variance with repeated measures was used to compare RER differences before and after the intervention and between groups. A t -test analysis was used to compare relative $\dot{V}O_{2\max}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) between groups. Table data are presented as means $\pm$ SD. An alpha level of 5% or less was set for all comparisons to discern statistical significance.

Intraclass correlation coefficients (ICCs) were performed to determine the reliability of the observed differences on ramp-incremental test variables pre-ride and post-ride $\dot{V}O_{2\max}$ and associated peak PO as well as the pre-ride to post-ride 3-minute all-out variables prepeak and postpeak PO, CP, and W' . Finally, Hedge's g statistic was used to measure the effect sizes when there were significant differences between means. The Bonferroni post hoc test was used to control for the family-wise error rate. Hedge's g effect size ranges for significant differences ($p \leq 0.05$) were calculated as follows: small ≥ 0.2 , medium ≥ 0.5 , and large ≥ 0.8 (Tables 1–3).

Results

The RIDE group completed the $90 \pm 8 \text{ km}$ (79–100 km) cycling route in 3 hours 12 minutes $\pm$ 4 minutes (2 hours 57 minutes–3 hours 21 minutes) at an average speed of $26 \pm 4 \text{ km} \cdot \text{h}^{-1}$. The variability within the distance traveled resulted from several subjects taking wrong turns. Within that period, HR fluctuated about a mean of $141 \pm 13 \text{ bpm}$. The proportion of time spent within the moderate-intensity, heavy-intensity, and severe-intensity domains based on the RAMP HR was $59 \pm 24\%$, $40 \pm 24\%$, and $1 \pm 1\%$, respectively (Figure 1 shows representative HR profile over the RIDE). Carbohydrate and fat intake at breakfast and during the long-duration ride of the RIDE group and the rest period of the CON group were similar. However, protein intake was reduced during the long-duration cycle for the RIDE group compared with the rest period of the CON group (Table 3).

Ramp-Incremental Exercise

In the CON group, $\dot{V}O_{2\max}$, peak $[La^-]$, peak aerobic power, peak HR, $\dot{V}O_2$ at GET, HR at GET, PO at GET, $\dot{V}O_2$ at RCP, HR at RCP, and PO at RCP (W) were not altered after 3 hours of sedentary rest (Table 1 and Figure 2). Moreover, absolute $\dot{V}O_{2\max}$ and associated peak aerobic power (W) were similar between the CON and RIDE groups at pre: $p = 0.14$ and post: $p = 0.469$, and pre: $p = 0.109$ and post: $p = 0.922$, respectively.

Compared with pre-90-km cycling, post-90-km RIDE subjects exhibited a reduction in both $\dot{V}O_{2\max}$ and associated PO and

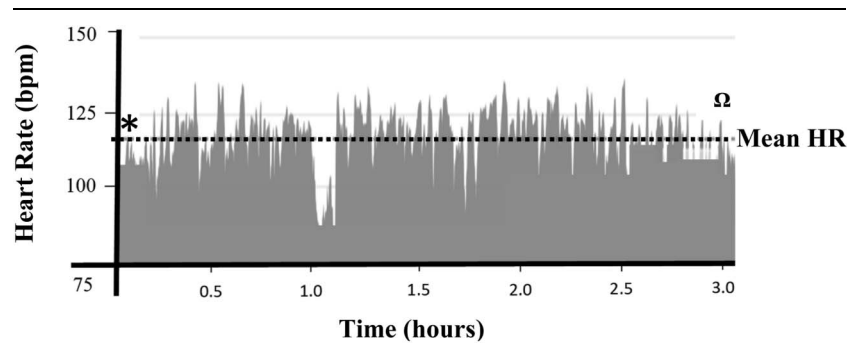


Figure 1. Representative heart rate (HR) (bpm, $b \cdot \min^{-1}$) profile over the 90-km long-duration ride for 1 subject. The asterisks denote the HR at the gas exchange threshold from the incremental tests for this subject, pre-90 km (*) and post-90 km (□). The dotted line is the mean HR over the long-duration cycle.

decreased $\dot{V}O_2$ at RCP ($p = 0.040$; Table 1 and Figure 3). No pre-90-km vs. post-90-km cycling changes were observed for the $\dot{V}O_2$ at GET ($p = 0.388$); however, the PO at GET was lower ($p = 0.027$; Table 1 and Figure 3). Respiratory exchange ratio was lower pre-90-km to post-90-km cycling from rest to just below RCP (Figure 4).

3-Minute All-Out Test

In the CON group, no differences in any other variables were observed after 3 hours of sedentary rest (Table 2 and Figure 2). In the RIDE group, W' decreased pre-90-km to post-90-km cycling by 32% ($p = 0.002$; Table 2 and Figure 3), whereas CP and peak power were unchanged ($p = 0.275$ and $p = 0.667$, respectively; Table 2 and Figure 3). Note that HR includes 8 of the 10 subjects because no HR data were collected for 2 of the subjects due to technical issues.

The ICCs on ramp-incremental test variables pre-ride and post-ride $\dot{V}O_{2\max}$ and associated peak PO showed 0.73 and 0.73, respectively, whereas pre-ride to post-ride 3-minute all-out variables prepeak and postpeak PO, CP, and W' resulted in ICCs of 0.82, 0.80, and 0.75, respectively. For the CON group, the ICCs performed before and after the ramp-incremental test for $\dot{V}O_{2\max}$

and PO were 0.92 and 0.94, respectively, whereas ICCs calculated on the 3-minute all-out test for peak PO, CP, and W' were 0.82, 0.80, and 0.75, respectively. All ICCs showed strong reliability preintervention to postintervention. Finally, the Hedge's g statistic reflecting effect sizes is presented in Tables 1, 2, and 3.

Discussion

This interventional study examined whether ramp-incremental and 3-minute all-out exercise-derived physiological parameters of intensity and associated POs were altered immediately after ~ 3 hours of road cycling. Our main findings pre-90 km to post-90 km of outdoor cycling showed (a) reductions of $\dot{V}O_{2\max}$ and RCP and associated POs; (b) decreased peak $[La^-]$; (c) lower POs at GET at the same $\dot{V}O_2$, on the pre-ramp- to post-ramp-incremental tests; and (d) unchanged CP and peak power, with decreased W' and peak $[La^-]$ pre-3-minute to post-3-minute all-out test. These data demonstrate that 3 hours of outdoor predominantly moderate-intensity cycling influence some, but not all, laboratory and physiological indices of exercise intensity domains and performance.

To ensure that the exercise protocols were sufficiently sensitive to discriminate pre-90-km vs. post-90-km cycling changes, an

Table 1

Data from the incremental test for the control group (CON) ($n = 6$) and 90-km cycling ride group (RIDE) ($n = 10$) with the exception of heart rate ($n = 8$).^{*†}

Variable	CON			RIDE			Hedges' g
	PRE (95% CI)	POST (95% CI)	p	PRE (95% CI)	POST (95% CI)	p	
$\dot{V}O_{2\max}$ ($L \cdot \min^{-1}$)	3.67 ± 0.43 (0.46)	3.66 ± 0.32 (0.34)	0.887	4.04 ± 0.48 (0.34)	3.80 ± 0.38 (0.27)	0.026†	0.50
$\dot{V}O_{2\max}$ ($ml \cdot kg^{-1} \cdot \min^{-1}$)	47.1 ± 2.6	47.2 ± 3.0	0.990	53.6 ± 4.0	50.4 ± 3.1	0.025†	0.98
HR peak ($b \cdot \min^{-1}$)	176 ± 9	176 ± 11.0	0.903	177 ± 11	176 ± 11	0.303	
PO_{peak} (W)	359 ± 46 (48)	353 ± 49 (51)	0.350	392 ± 30 (22)	357 ± 26 (19)	0.002†	1.14
Baseline $[La^-]$ (mM)	1.8 ± 0.3 (0.32)	2.7 ± 0.5 (0.48)	0.002†	2.3 ± 0.4 (0.31)	2.8 ± 0.8 (0.54)	0.139	
Peak $[La^-]$ (mM)	13.4 ± 1.6 (1.6)	12.7 ± 3.2 (1.6)	0.545	11.4 ± 1.6 (1.16)	8.4 ± 3.6 (2.54)	0.015†	0.98
$\dot{V}O_2$ at RCP ($L \cdot \min^{-1}$)	3.15 ± 0.30 (0.31)	3.14 ± 0.27 (0.29)	0.818	3.49 ± 0.46 (0.33)	3.34 ± 0.43 (0.31)	0.040†	0.30
HR at RCP ($b \cdot \min^{-1}$)	161 ± 6	163 ± 8	0.521	165 ± 12	164 ± 13	0.866	
PO at RCP (W)	259 ± 30 (31)	255 ± 33 (34)	0.346	292 ± 40 (29)	272 ± 24 (17)	0.023†	0.55
$\dot{V}O_2$ at GET ($L \cdot \min^{-1}$)	2.29 ± 0.21 (0.22)	2.32 ± 0.20 (0.21)	0.437	2.52 ± 0.44 (0.32)	2.44 ± 0.38 (0.27)	0.388	
HR at GET ($b \cdot \min^{-1}$)	139 ± 7	144 ± 9	0.067	139 ± 13	141 ± 11	0.661	
PO at GET (W)	168 ± 23 (25)	167 ± 27 (28)	0.858	194 ± 32 (22)	178 ± 25 (18)	0.027†	0.51

^{*}CI = confidence interval; $\dot{V}O_{2\max}$ = maximal oxygen consumption; HR peak = peak heart rate ($b \cdot \min^{-1}$); PO_{peak} = power output associated with $\dot{V}O_{2\max}$; Baseline $[La^-]$ = blood lactate concentration measured before the ramp-incremental test; Peak $[La^-]$ = blood lactate concentration measure after the ramp-incremental test; $\dot{V}O_2$ = oxygen consumption; RCP = respiratory compensation point; PO = power output; GET = gas exchange threshold.

[†]Values are reported as mean $\pm$ SD. Effect sizes (Hedges' g) and 95% CIs are calculated on those comparisons that were $p \leq 0.05$.

‡ $p \leq 0.05$ preintervention to postintervention.

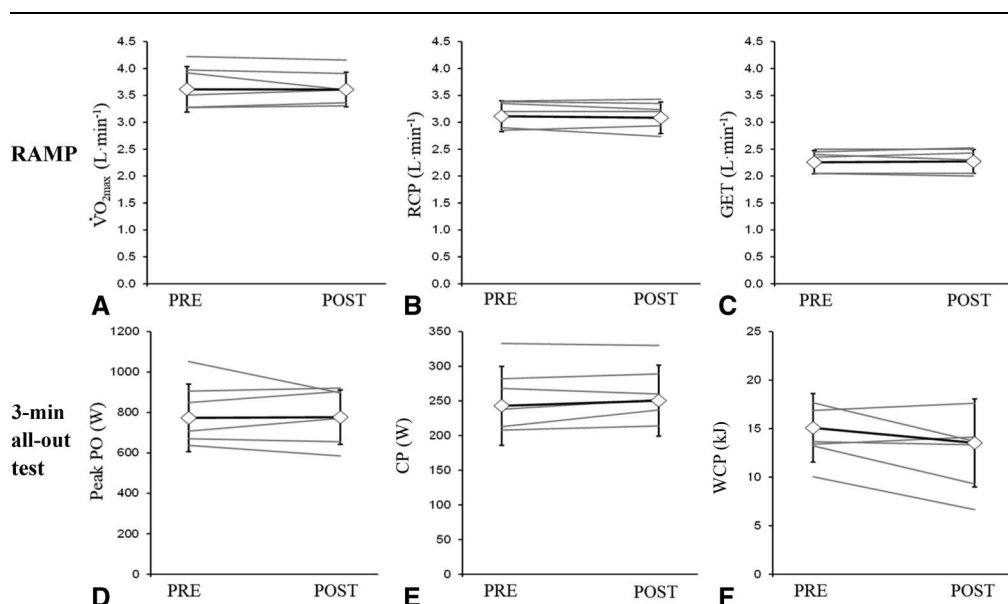


Figure 2. Mean and individual results for the CON group ($n = 6$) PRE and POST the 3-hour sedentary period. Panels A–C were derived from the ramp-incremental exercise (RAMP): (A) $\dot{V}O_{2\max}$ (L·min⁻¹), (B) RCP (L·min⁻¹), and (C) GET (L·min⁻¹) estimated lactate threshold. Panels D–F were derived from the 3-minute all-out tests: (D) Peak PO (W), (E) CP (W), and (F) WCP (kJ). Open diamond values are mean ± SD. Solid lines indicate changes PRE- vs. POST-sedentary period. Gray lines indicate individual PRE- to POST-sedentary period. $\dot{V}O_{2\max}$ = maximal oxygen uptake; RCP = respiratory compensation point; Peak PO = peak power output; CP = critical power; WCP = work performed above CP; GET = gas exchange threshold.

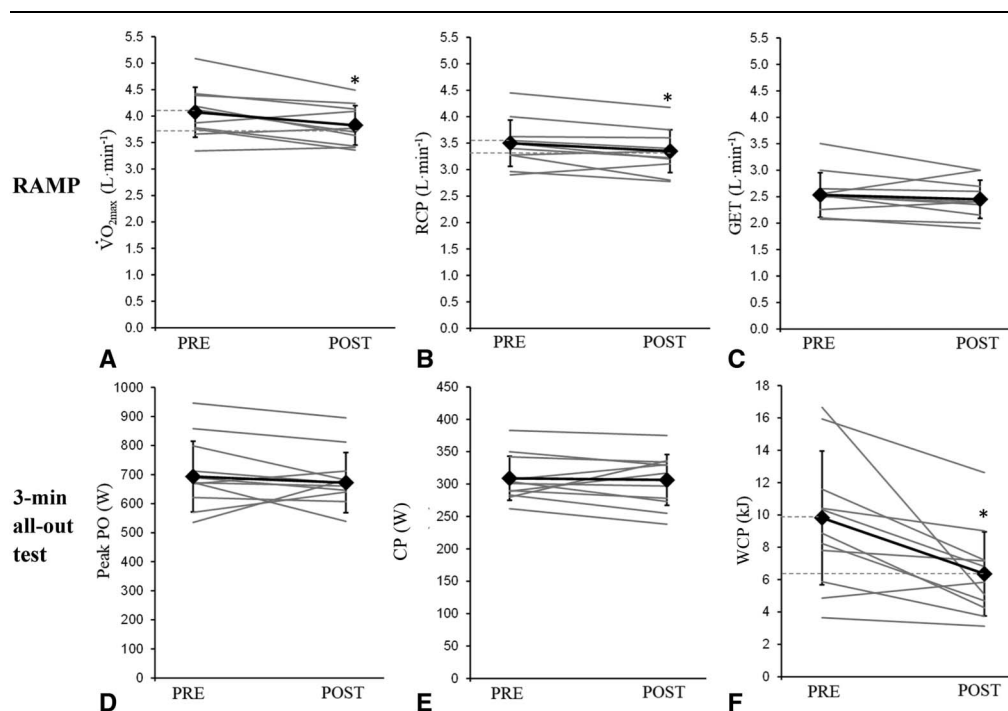


Figure 3. Mean and individual results for the RIDE group ($n = 10$) PRE and POST the long-duration cycle. Panels A–C were derived from the ramp-incremental exercise (RAMP): (A) $\dot{V}O_{2\max}$ (L·min⁻¹), (B) RCP (L·min⁻¹), and (C) GET (L·min⁻¹) estimated lactate threshold. Panels D–F were derived from the 3-minute all-out test: (D) Peak PO (W), (E) CP (W), and (F) WCP (kJ). Closed diamond values are reported as mean ± SD. Solid lines indicate changes PRE- vs. POST-90-km cycling. Gray lines indicate individual PRE- to POST-90-km cycling. * PREintervention to POSTintervention difference ($p < 0.05$). Table 1 shows effect sizes of significant differences. $\dot{V}O_{2\max}$ = maximal oxygen uptake; RCP = respiratory compensation point; Peak PO = peak power output; CP = critical power; WCP = work performed above CP; GET = gas exchange threshold.

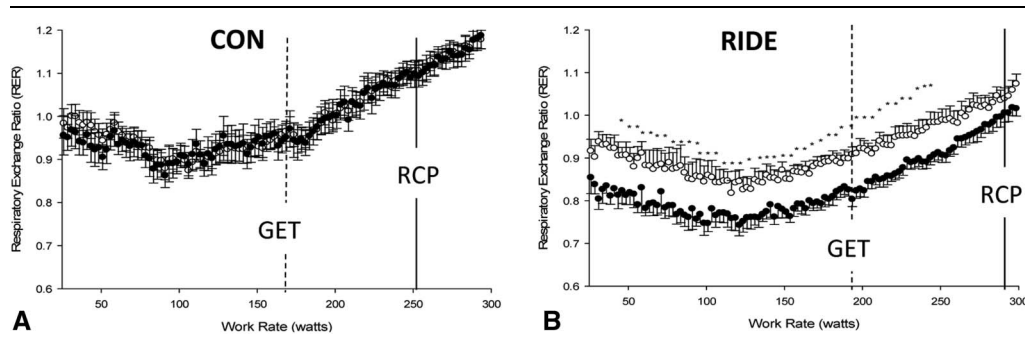


Figure 4. A) Control group (CON) ramp-incremental exercise (RAMP) RER (respiratory exchange ratio), 5 seconds averaged data (mean $\pm$ SE). B) Ride group (RIDE) RAMP RER, 5 seconds averaged data (mean $\pm$ SE). Open circles indicate PREintervention, closed circles indicate POSTintervention, dashed vertical line indicates GET estimated lactate threshold, and solid vertical line indicates RCP. *PREintervention to POSTintervention difference ($p < 0.05$). GET = gas exchange threshold.

independent group of healthy active control subjects performed the ramp-incremental test and the 3-minute all-out exercise before and after a sedentary period matched to the 90-km cycle completed by the RIDE group. Within this control group, neither $\dot{V}O_{2max}$ and associated PO nor $\dot{V}O_2$ and PO associated with GET and RCP, as well as peak $[La^-]$, differed after 3 hours of seated rest during the ramp-incremental exercise. Similarly, for the 3-minute all-out cycling test, the CON group exhibited unchanged W' , CP, peak PO, $\dot{V}O_{2max}$ and associated PO, and baseline and peak $[La^-]$. Collectively, these data suggest that the observed changes in the RIDE group stemmed from the long-duration ride and not the exercise protocols completed ~ 3 hours earlier because the ICC data showed test-retest reliability as good to excellent.

To the best of our knowledge, no previous study has investigated the physiological and PO responses to ramp-incremental exercise of long-duration cycling exercise. The pre-RIDE to post-ride ramp-incremental exercise novel observation that GET was identified at a similar $\dot{V}O_2$ but at a reduced PO presents 2 possibilities. First, the $\dot{V}O_2$ kinetics of the subjects in the RIDE group may have been slowed, so the “gap” between $\dot{V}O_2$ and the ramp-derived PO within the moderate-intensity range of the ramp was widened (9,20). Second, if $\dot{V}O_2$ kinetics remained unchanged, a reduction in work efficiency (i.e., the same $\dot{V}O_2$ supporting a lower PO) underpinned this result (34). The greater O₂ cost associated with fat oxidation would explain this

observation (7) because during the post-ride ramp-incremental test, RER was consistently lower from baseline to above GET (Figure 4) (36). Although both hypotheses are tenable, the latter might be anticipated to predominate in scenarios where glycogen depletion is evident such as the completion of a 3-hour cycle ride (11,43). This direct association between glycogen stores and predominantly aerobically based long-duration performances has been a consistent finding in the literature (3).

Moreover, this reduced PO at GET, that defines the upper limit of the moderate-intensity domain, after or during the latter portion of a long-duration moderate-intensity to heavy-intensity exercise is of practical importance as performing the same PO later in the ride would require a higher $\dot{V}O_2$ and a transition from the moderate-intensity to the heavy-intensity domain (5,14,32). In addition, this change in intensity domain would also result in greater glycogen depletion resulting in a decreased time to fatigue (4,28,43).

By contrast, previous research studying the effects of priming ramp-incremental exercise on a subsequent and identical incremental exercise observed a higher O₂ cost for the same power outputs performed during the priming exercise (5). It was suggested that the improved blood flow and reduced type II fiber recruitment after priming exercise would elicit this response (5). This study’s differing results after priming exercise (90-km cycle ride) suggests that the priming exercise’s reduced intensity and increased duration and O₂ cost associated with increased fat oxidation (7) have underpinned this change in efficiency.

Table 2

Data from the 3-min all-out test ($n = 6$) for the control group (CON), $n = 10$ for all non-HR data in the long-duration cycle group (RIDE), and $n = 8$ for all HR data in the RIDE group.*†

Variable	CON			RIDE			Hedges' <i>g</i>
	PRE	POST	<i>p</i>	PRE (95% CI)	POST (95% CI)	<i>p</i>	
$\dot{V}O_{2peak}$ (L·min ⁻¹)	3.58 $\pm$ 0.51	3.53 $\pm$ 0.41	0.373	3.85 $\pm$ 0.30	3.70 $\pm$ 0.45	0.158	
Peak HR (b·min ⁻¹)	172 $\pm$ 8	173 $\pm$ 7	0.137	168 $\pm$ 10	170 $\pm$ 12	0.813	
Baseline $[La^-]$ (mM)	9.3 $\pm$ 1.1	8.3 $\pm$ 1.5	0.217	6.6 $\pm$ 1.8 (1.1)	4.5 $\pm$ 2.1 (1.3)	0.022‡	0.98
Peak $[La^-]$ (mM)	17.9 $\pm$ 1.5	15.9 $\pm$ 3.2	0.265	14.9 $\pm$ 2.4 (1.5)	10.1 $\pm$ 2.4 (1.5)	0.0009‡	1.83
Peak PO (W)	761 $\pm$ 180	756 $\pm$ 134	0.898	697 $\pm$ 135	687 $\pm$ 105	0.667	
WCP (kJ)	14.8 $\pm$ 3.8	12.8 $\pm$ 4.6	0.071	9.8 $\pm$ 3.4 (2.1)	6.8 $\pm$ 2.6 (2.1)	0.002‡	0.91
EP (W)	248 $\pm$ 61	253 $\pm$ 56	0.240	304 $\pm$ 26	297 $\pm$ 34	0.275	

*CI = confidence interval; $\dot{V}O_{2peak}$ = peak oxygen consumption; Peak HR = peak heart rate at volitional fatigue on the $\dot{V}O_{2max}$ test; Baseline $[La^-]$ = blood lactate concentration measured before the 3-min all-out test; Peak $[La^-]$ = blood lactate concentration measured after the 3-min all-out test; Peak PO = peak power output during the 3-minute all-out test; WCP = estimate of the total amount of work that can be performed above end-test PO; EP = end-test PO.

†Values are reported as mean $\pm$ SD. 95% CIs calculated for statistically significant comparisons. Effect sizes (Hedges' *g*) and 95% CIs are calculated on those comparisons that were $p \leq 0.05$.

‡ $p \leq 0.05$ preintervention to postintervention.

Both the mean $\dot{V}O_2$ and PO at RCP were reduced after 90 km of outdoor cycling. The RCP approximates the metabolic rate at the heavy-intensity to severe-intensity domain boundary (21) and has been linked to the upper limit of sustainable performance (20). This reduced PO at RCP after the 90-km outdoor ride was concomitant with a reduction in $\dot{V}O_2$ ($\sim 10 \text{ ml} \cdot \text{min}^{-1} \cdot \text{W}^{-1}$; $20 \text{ W} = \sim 200 \text{ ml O}_2$), which, unlike GET, implies that work efficiency may have been preserved above the moderate-intensity domain as a consequence of carbohydrates becoming the predominant substrate at supralactate threshold intensities (5,43). This change to glycolytic phosphorylation from fat oxidation, as observed through the reduced RER after a 90-km cycle ride as the predominant energy system, would account for the reduced $\dot{V}O_2$ per ADP phosphorylated (36). However, the observed reduction in the PO at the RCP would have performance-related decrements after long-duration exercise (14,16,23).

Reductions in $\dot{V}O_{2\max}$ and associated PO were also observed in the RIDE group after 90 km of outdoor cycling. It follows that the marked decrease in $\dot{V}O_{2\max}$ after several hours of cycling, would be expressed during those efforts at which $\dot{V}O_{2\max}$ is attained (18). The decreased $\dot{V}O_{2\max}$ and associated PO may be a consequence of the reduced glycogen stores, which would limit not only maximal performance but also the duration of continuous exercise performed at critical power (11,28). Moreover, recent research on the impact of reduced glycogen stores on cycling performances at intensities $>85\%$ of $\dot{V}O_{2\max}$ has shown reduced maximal aerobic power (3,40).

In contrast to our hypothesis, mean CP, which represents an approximation of the boundary of the upper limit of sustainable exercise (8,41), did not change after the 90-km ride (Table 2). A similar conservation of constant power performance after exercise has also been observed by Miura et al. (28). Their results were observed after completing a glycogen depletion protocol consisting of 75 minutes of cycling at $60\% \dot{V}O_{2\max}$, followed by repeated high-intensity 1-minute cycling bouts. The findings of Miura et al. (28) and the present study differ from Clark et al. (11) who observed a decline in CP of trained cyclists after 2 hours of laboratory-based constant-power exercise performed within the heavy-intensity domain. Despite study similarities, fundamental differences exist between the experiments of Clark et al. (11) and the present study, i.e., the increased duration of the exercise of the present study (3 vs. 2 hours), the inherent characteristics of the cycling exercise (i.e., the present study, outdoor, variable PO cycling vs. indoor, constant-power exercise), and the intensity of exercise (i.e., $\sim 60\%$ in the moderate-intensity domain in the present study vs. 100% within the heavy-intensity domain exercise in Clark et al. (11)). From a physiological perspective, the divergent CP study findings, again, may stem from the magnitude of glycogen depletion. Despite exercising for a shorter duration, Clark et al. (11) used a higher exercise intensity that might have elicited greater glycogen use (13) compared with the present study, where most of the cycling time was spent within the moderate-intensity domain. Moreover, in the study by Clark et al. (11), no carbohydrates (CHO) were available to the subjects, whereas, in the current study, the mean consumption of CHO of the cyclists during the long-duration cycle was 132 grams (Table 3). CHO ingestion during long-duration exercise has been shown to facilitate improved performances and reduced muscle glycogen use after long-duration exercise (15,17). Such muscle glycogen sparing in the present study would be anticipated to contribute to preserving CP (10,43).

Finally, the RIDE group of the current study exhibited a W' reduction from before to after long-duration cycle as

Table 3

Carbohydrate, protein, and fat intake at breakfast and during the 90-km cycle for the RIDE group and nonexercising for controls (CON).*

Breakfast	CON (95% CI)	RIDE (95% CI)	<i>p</i>	Hedges' <i>g</i>
Carbohydrates (g)	66 ± 59	54 ± 59	0.68	
Proteins (g)	20 ± 12	13 ± 9	0.19	
Fats (g)	16 ± 6	8 ± 6.1	0.13	

Intervention period	CON	RIDE	<i>p</i>	
Carbohydrates (g)	154 ± 29	132 ± 71	0.44	
Proteins (g)	19 ± 4 (3.2)	11 ± 8 (6.4)	0.03†	1.10
Fats (g)	10 ± 2	9 ± 7	0.60	

*Effect sizes (Hedges' *g*) and 95% confidence intervals (95% CI) are calculated on those comparisons that were $p \leq 0.05$.

†Significant differences $p < 0.05$.

hypothesized. However, the decrease in W' was greater (32%) in the study by Clark et al. (11) compared with the current study (20%). This difference could be attributed to the increased duration in the heavy-intensity domain in the study by Clark et al. (11) compared with the current investigation, leading to a greater substrate reduction for anaerobic glycolytic phosphorylation of type I and type II fibers reducing PO (43). Furthermore, previous research has demonstrated that exercising at $> \text{GET}$ exercise intensities compared with $< \text{GET}$ increases glycogen use in type II muscle fibers (43). Furthermore, the decreased $[\text{La}^-]$ observed in post-ride 3-minute all-out performance in the present study reflects reduced anaerobic glycolytic phosphorylation that underpins the observed decrease of W' (39). Finally, the unchanged peak power during the 3-minute all-out test, pre-to post long-duration ride, has also been shown in earlier work after a 2-hour heavy-intensity ride performed under controlled conditions. This agrees with previous literature that suggests that peak power, which is buttressed by ATP hydrolysis, is not affected by glycogen availability (11,37,42).

An interesting observation from our data is that the expression of the acute performance decrements depends on the tests used. For example, if only peak and critical power were tested in the present study, we would conclude that the 3-hour outdoor cycling had minimal effects on performance. However, our conclusions are different with the inclusion of metabolic and performance measures evaluating GET, RCP, $\dot{V}O_{2\max}$, CP, W' , and blood lactate concentrations. Thus, although acute performance decrements show promise as a tool for monitoring training load (31), interpretation of the results should be limited to the performance and physiological variables associated with those specific results.

This is the first study to observe the changes to $\dot{V}O_2$ and PO at GET, RCP, and $\dot{V}O_{2\max}$ pre-long-duration to post-long-duration cycling. Moreover, incorporating the long-duration cycle on the open road gives a real-world element to the results that cannot be achieved through a laboratory-based cycle underpinned by an unchanging PO. However, the use of HR as a measure to demarcate exercise intensity during the 90-km ride is a limitation because HR is affected by numerous factors, such as temperature, hydration, and duration of exercise (1), which could affect measures of this variable during exercise (1).

Not using the RIDE group as their own control was also a limitation; however, the CON group showed very reproducible results, and we would anticipate that due to the nature of the RIDE subjects' training, which often requires exhaustive intense efforts, they would likely exhibit even higher reproducibility.

Finally, we would not expect any meaningful differences associated with the efficacy of the performances of the CON group because these subjects had all performed several RAMP tests and 3-minute all-out trials with other studies requiring similar testing procedures that had been undertaken in our laboratory. Furthermore, the 5-minute time between the 3-hour ride and post-ride testing would have resulted in significant energy reconstitution and metabolic byproduct clearance at rest post-ride (Table 1) and affected the performance measures. Finally, having no measure of PO during the ride was also a methodological shortcoming.

Practical Applications

Recreational cyclists performing a long-duration ride at intensities mostly below GET should expect a reduction in PO at intensities at $\geq$ GET. This would lead to premature fatigue after long-duration exercise if attempting to maintain a PO equivalent to their initial lactate threshold or higher. However, the preserved CP preintervention to postintervention in the RIDE group demonstrates that no decline in CP performances after approximately 3 hours of predominantly moderate-intensity exercise should be expected. However, the observed decrease in W' after long-duration ride that sustained POs above CP would be impaired. Such changes in performance should be considered when developing pacing strategies for longer-duration competitive cycling or triathlon events, as well as long-duration training sessions in moderately trained cyclists. The study has shown that performing a ramp-incremental and 3-minute all-out test, followed by a long-duration ride at a predominantly moderate-intensity, reduces PO at GET without a change in $\dot{V}O_2$ on a similar test after cycling. Moreover, reductions in RCP and $\dot{V}O_{2max}$, and accompanying POs, after cycling on the ramp-incremental test were observed. Finally, CP was preserved preintervention to postintervention in the RIDE group in the present study.

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